

A sum concentrated at its last few terms

$\sum (k/n)^k$ and the geometric series with ratio $1/e$

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \left(\frac{k}{n}\right)^k = \sum_{j=0}^{\infty} e^{-j} = \frac{e}{e-1} = 1.5819767 \dots$$

Overview. Count from the top. Writing $j = n - k$, the terms become

$$a_{n,j} = \left(1 - \frac{j}{n}\right)^{n-j},$$

and for each fixed j ,

$$a_{n,j} = \left(1 - \frac{j}{n}\right)^n \left(1 - \frac{j}{n}\right)^{-j} \longrightarrow e^{-j} \cdot 1.$$

So the last few terms of the sum converge to the geometric sequence $1, e^{-1}, e^{-2}, \dots$, whose total is $\frac{e}{e-1}$. All the work consists of showing that everything else, the first few terms and the entire middle, contributes nothing in the limit, uniformly in n . Two crude geometric bounds suffice: $(k/n)^k \leq 2^{-k}$ for $k \leq n/2$ and $(k/n)^k \leq e^{-j/2}$ for $j \leq n/2$.

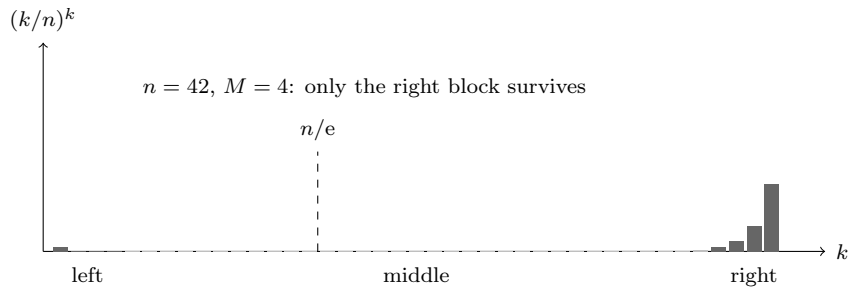
1. Setting up a three-block split

Let

$$S_n = \sum_{k=1}^n \left(\frac{k}{n}\right)^k.$$

Fix an integer $M \geq 1$ and take $n > 2M$. Substituting $j = n - k$ in the top block, split

$$S_n = \underbrace{\sum_{k=1}^M \left(\frac{k}{n}\right)^k}_{\text{left}} + \underbrace{\sum_{k=M+1}^{n-M-1} \left(\frac{k}{n}\right)^k}_{\text{middle}} + \underbrace{\sum_{j=0}^M a_{n,j}}_{\text{right}}, \quad a_{n,j} = \left(1 - \frac{j}{n}\right)^{n-j}. \quad (1)$$



2. The right block converges term by term

For j fixed and $n \rightarrow \infty$,

$$\left(1 - \frac{j}{n}\right)^n \longrightarrow e^{-j}, \quad \left(1 - \frac{j}{n}\right)^{-j} \longrightarrow 1,$$

so $a_{n,j} \rightarrow e^{-j}$. The right block has a fixed number $M + 1$ of terms, so

$$\lim_{n \rightarrow \infty} \sum_{j=0}^M a_{n,j} = \sum_{j=0}^M e^{-j}. \quad (2)$$

The tail of this geometric series is

$$\sum_{j>M} e^{-j} = \frac{e^{-M}}{e-1} \cdot e,$$

which is small for large M .

3. The left block is $O(M^2/n)$

For $1 \leq k \leq M$ we have $\frac{k}{n} \leq 1$, hence $(k/n)^k \leq \frac{k}{n}$, and

$$0 \leq \sum_{k=1}^M \left(\frac{k}{n}\right)^k \leq \frac{1}{n} \sum_{k=1}^M k = \frac{M(M+1)}{2n} \leq \frac{M^2}{n}. \quad (3)$$

For each fixed M this tends to 0 as $n \rightarrow \infty$. Note the order of quantifiers: M is frozen first, $n \rightarrow \infty$ second, and only at the very end is M sent to infinity.

4. The middle block is uniformly small

This is the only estimate that must hold uniformly in n . Split the middle at $k = n/2$.

Lower half, $M < k \leq n/2$. Here $\frac{k}{n} \leq \frac{1}{2}$, so

$$\left(\frac{k}{n}\right)^k \leq \left(\frac{1}{2}\right)^k = 2^{-k}, \quad \text{hence} \quad \sum_{M < k \leq n/2} \left(\frac{k}{n}\right)^k \leq \sum_{k>M} 2^{-k} = 2^{-M}. \quad (4)$$

Upper half, $M < j < n/2$, **where** $j = n - k$. Using $\log(1-x) \leq -x$,

$$\left(\frac{k}{n}\right)^k = \left(1 - \frac{j}{n}\right)^{n-j} = \exp\left[(n-j) \log\left(1 - \frac{j}{n}\right)\right] \leq \exp\left[-\frac{j(n-j)}{n}\right] \leq e^{-j/2}, \quad (5)$$

because $j < \frac{n}{2}$ gives $n-j > \frac{n}{2}$ and therefore $\frac{j(n-j)}{n} > \frac{j}{2}$. Summing,

$$\sum_{M < j < n/2} \left(\frac{k}{n}\right)^k \leq \sum_{j>M} e^{-j/2} = \frac{e^{-(M+1)/2}}{1 - e^{-1/2}}. \quad (6)$$

Both (4) and (6) are independent of n , and both tend to 0 as $M \rightarrow \infty$. This is exactly what makes the interchange of limits legitimate.

5. Conclusion

Write $L = \sum_{j \geq 0} e^{-j} = \frac{e}{e-1}$. By (1),

$$|S_n - L| \leq \underbrace{\frac{M^2}{n}}_{\text{left, (3)}} + \underbrace{2^{-M} + \frac{e^{-(M+1)/2}}{1 - e^{-1/2}}}_{\text{middle, (4),(6)}} + \underbrace{\left| \sum_{j=0}^M a_{n,j} - \sum_{j=0}^M e^{-j} \right|}_{\text{tail}} + \underbrace{\sum_{j>M} e^{-j}}_{\text{tail}}.$$

Fix $\varepsilon > 0$. Choose M so large that the middle bound and the tail together are $< \frac{\varepsilon}{2}$; this is possible by (4), (6) and the convergence of $\sum e^{-j}$, and the choice does not depend on n . With M now fixed, choose N so large that for $n \geq N$ both $\frac{M^2}{n}$ and the finite-sum discrepancy (which tends to 0 by (2)) are together $< \frac{\varepsilon}{2}$. Then $|S_n - L| < \varepsilon$ for $n \geq N$, so

$$\boxed{\lim_{n \rightarrow \infty} \sum_{k=1}^n \left(\frac{k}{n}\right)^k = \frac{e}{e-1}}.$$

6. The refined asymptotics

The same expansion gives the next order. For fixed j ,

$$\log a_{n,j} = (n-j) \log \left(1 - \frac{j}{n}\right) = (n-j) \left(-\frac{j}{n} - \frac{j^2}{2n^2} - \dots\right) = -j + \frac{j^2}{2n} + O\left(\frac{j^3}{n^2}\right),$$

so $a_{n,j} = e^{-j} \left(1 + \frac{j^2}{2n} + \dots\right)$. Adding the left block, whose leading term is $\frac{1}{n}$ from $k=1$, one predicts

$$S_n = \frac{e}{e-1} + \frac{c}{n} + o\left(\frac{1}{n}\right), \quad c = 1 + \frac{1}{2} \sum_{j \geq 0} j^2 e^{-j} = 1 + \frac{e(e+1)}{2(e-1)^3}, \quad (7)$$

using $\sum_{j \geq 0} j^2 x^j = \frac{x(1+x)}{(1-x)^3}$ at $x = e^{-1}$. Numerically $c = 1.996142\dots$, close to but not equal to 2.

n	S_n	$n \left(S_n - \frac{e}{e-1}\right)$
10	1.9080529	3.26076
100	1.6027982	2.08215
1000	1.5839809	2.00420
10000	1.5821764	1.99695
∞	1.5819767	1.99614 = c

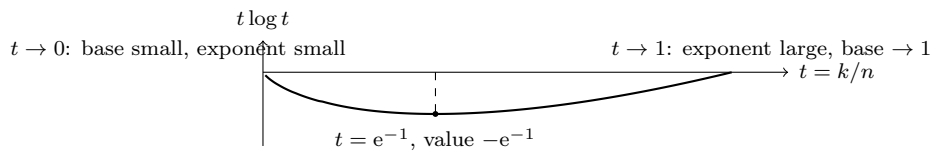
The numerics confirm (7): the rescaled errors descend towards $c = 1.9961$, the approach itself being $O(1/n)$.

7. Where the mass really is

It is worth locating the maximum of the terms, because it explains why counting from the top is the right idea and counting from the bottom is not. Writing $t = k/n$,

$$\log \left(\frac{k}{n}\right)^k = k \log \frac{k}{n} = n t \log t,$$

and $t \log t$ is minimised at $t = e^{-1}$ with value $-e^{-1}$. So the *smallest* term sits near $k = n/e$, where it is of size $e^{-n/e}$, exponentially negligible, and the terms grow monotonically towards both ends. At the left end they are small because k is small; at the right end $(k/n)^k$ approaches 1. Hence the sum is a two-sided affair with a vanishing valley in between, and only the right end carries mass of order 1.



The two ends are not symmetric. At the left end the exponent k is small but so is the base, and $(k/n)^k \leq \frac{k}{n} \rightarrow 0$: each term individually vanishes. At the right end the base tends to 1 while the exponent grows like n , and the product $(n-j) \log(1 - \frac{j}{n})$ stabilises at $-j$: the terms survive. That asymmetry is what makes (3) cheap and (5) essential, and it is the reason the limit is a geometric series in e^{-1} rather than anything involving the valley at $k = n/e$.